

# Medial-axis reconstruction in positive curvature: the focal collapse and total reconstruction\*

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## Abstract

For a closed set  $X$  in Euclidean or Hadamard space the medial-axis reconstruction  $\mathcal{R}_X$  is a strict subset of the complement of  $X$ : Białożyty’s Theorem 6 [1] describes it as the convex (horoball) hull together with the interiors of the *defective* supporting half-spaces (horoballs) — those whose contact set is non-convex. We show that in *positive* curvature this entire structure collapses. On the round sphere  $\mathbb{S}^n$  the reconstruction is *total*: for every closed  $X \subseteq \mathbb{S}^n$  with at least two points,

$$\mathcal{R}_X = \mathbb{S}^n \setminus X$$

(Theorem 4.1). The mechanism is a *focal* one. A free geodesic ball that touches  $X$  at two points has, at its centre, the full power of a circumcentre: in particular the pole of any free hemisphere whose equator meets  $X$  twice reconstructs the whole hemisphere, *irrespective of the convexity of the contact set* (Lemma 3.1), so the Motzkin/non-convexity criterion that governs the Euclidean and hyperbolic theories is simply bypassed. More fundamentally, compactness of geodesics replaces the Euclidean “escape to infinity”: following the ray from the nearest contact of any point  $p \notin X$ , a second nearest contact must appear before the antipode, producing a medial centre that swallows  $p$  (Theorem 4.1). The Białożyty certificate then strictly undercounts  $\mathcal{R}_X$  — we exhibit a *convex* set (a spherical cap) whose whole complement is reconstructed, and a circle whose far hemisphere is reconstructed although it has no defective supporting hemisphere (§5). We then ask what survives off constant curvature. Total reconstruction is a *constant-curvature* phenomenon: on a general compact manifold the cut locus can shield a point from every medial ball, but the obstruction is removed by a spread hypothesis — if  $X$  is  $\delta$ -dense with  $2\delta < \text{inj}(M)$  then  $\mathcal{R}_X = M \setminus X$  (Theorem 6.1), a condition made explicit under a curvature upper bound by Klingenberg’s estimate. Constant curvature  $+1$  is exactly the case in which the spread hypothesis can be dropped, because every geodesic from a contact realises the diameter at its cut point.

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### 1 Introduction

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\*Fourth note in a series: [2] generalised the sufficiency half of Theorem 6 of [1] to Hadamard manifolds; [3] proved the full characterisation in real hyperbolic space  $\mathbb{H}^n$ ; [4] isolated the Gauss obstruction and treated flat, CAT(0) and complex-hyperbolic horospheres. This note crosses into *positive* curvature, where the picture inverts. Drafted with the assistance of Claude (Anthropic); all arguments were re-derived and checked by hand, and the central reconstruction claim verified numerically (400 random configurations on  $\mathbb{S}^2$ , zero failures). Remaining errors are ours.

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## 1 Introduction

### The Euclidean and Hadamard theories, and what they rest on

For a closed set  $X$  in a complete Riemannian manifold  $M$  write  $M_X$  for its *medial axis* — the points with two or more nearest points of  $X$  — and, for  $a \in M_X$ ,  $r(a) = \text{dist}(a, X)$  for its reach. The *reconstruction* of  $X$  is the union of the open medial balls,

$$\mathcal{R}_X = \bigcup_{a \in M_X} B(a, r(a)).$$

Białożyty's Theorem 6 [1] characterises  $\mathcal{R}_X$  for closed  $X \subseteq \mathbb{R}^n$ :

$$\mathcal{R}_X = (\widehat{X} \setminus X) \cup \bigcup \{\text{int } H : H \text{ a supporting half-space, } X \cap \partial H \neq \text{conv}(X \cap \partial H)\}, \quad (1)$$

with  $\widehat{X} = \overline{\text{conv}} X$ ; the companion notes carry this to hyperbolic space [3] and, with horoballs in place of half-spaces and a horospherical convex hull in place of  $\text{conv}$ , to every Hadamard manifold [2, 4]. Two features are common to all of these *non-positively* curved theories.

First,  $\mathcal{R}_X$  is a *strict* subset of  $M \setminus X$ . The prototype is two points  $x_0, x_1 \in \mathbb{R}^2$ : the open segment between them (the hull) and the two open half-planes bounded by the line  $x_0x_1$  (the contact set  $\{x_0, x_1\}$  is non-convex) are reconstructed, but the two *rays* of that line beyond the segment are not — a point  $q$  on the ray past  $x_0$  has  $\text{dist}(a, q) > \text{dist}(a, x_0) = r(a)$  for every medial  $a$  (all of which lie on the perpendicular bisector), so  $q \notin \mathcal{R}_X$ . The unreconstructed points are exactly those from which the ascending flow of  $\text{dist}(\cdot, X)$  *escapes to infinity* with a single nearest point throughout.

Second, the detection of which supporting half-spaces contribute is a genuine *convexity* question — Motzkin's theorem (a closed set in  $\mathbb{R}^m$  is convex iff it is Chebyshev) is what turns “admits a medial centre” into “has a non-convex contact set.” In [4] this is exactly the step that becomes hard on curved horospheres.

## Positive curvature inverts both features

We show that on the round sphere  $\mathbb{S}^n$  *both* mechanisms disappear, and in the most decisive way.

The convexity question evaporates first. A free geodesic ball that meets  $X$  at its boundary in two points already has, at its *centre*, two nearest contacts — so the centre is medial, with no condition on the contact set whatsoever. The extreme case is a hemisphere: the pole of any free hemisphere is a circumcentre equidistant (at  $\frac{\pi}{2}$ ) from its entire equator, so if the equator meets  $X$  twice the pole reconstructs the whole hemisphere, *convex contact set or not* (Lemma 3.1). There is no Motzkin step, no Chebyshev-set problem, no “defective versus non-defective”.

The strictness disappears next, and globally. On  $\mathbb{S}^n$  a geodesic is a great circle; the ray from the nearest contact  $x_0$  of a point  $p \notin X$  cannot escape — it reaches the antipode  $-x_0$  at distance  $\pi$ , where  $x_0$  is the *farthest* point of  $X$ , so some second contact  $x_1$  has overtaken it strictly earlier. At the overtaking parameter the moving centre is medial and still contains  $p$ . Hence *every* point off  $X$  is reconstructed:

$$\mathcal{R}_X = \mathbb{S}^n \setminus X \quad (\text{any closed } X, |X| \geq 2).$$

This is Theorem 4.1. The Białożył certificate (1) is then not merely reformulated but *strictly undercounts*: in §5 we give a convex spherical cap whose complement is entirely reconstructed (the certificate predicts nothing), and a latitude circle whose far hemisphere is reconstructed although every supporting hemisphere touches it once (no defective hemisphere). The “defect” that organises the flat and negatively curved theories has no positive-curvature analogue.

## What survives off constant curvature

Total reconstruction turns out to be a feature of *constant* positive curvature, not of positivity alone. The proof on  $\mathbb{S}^n$  uses that every geodesic from  $x_0$  realises the diameter  $\pi$  at its cut point, so the nearest contact is dethroned before the geodesic stops minimising. On a general compact manifold the cut locus of  $x_0$  can occur *early* — a second minimising geodesic back to the same point  $x_0$ , with no rival contact yet in sight — and such a configuration can shield a point from every medial ball (§7). The shielding is removed by a quantitative spread hypothesis: if  $X$  is  $\delta$ -dense with  $2\delta < \text{inj}(M)$ , a rival contact is forced within distance  $2\delta$ , well inside the minimising range, and  $\mathcal{R}_X = M \setminus X$  again (Theorem 6.1). Under a curvature upper bound  $\text{sec} \leq \Delta$  the injectivity radius is controlled by Klingenberg’s estimate, making the hypothesis explicit. Thus:

| Curvature                                                                 | $X$                                        | $\mathcal{R}_X$                                                                               |
|---------------------------------------------------------------------------|--------------------------------------------|-----------------------------------------------------------------------------------------------|
| $\text{sec} \equiv 0$ ( $\mathbb{R}^n$ ) / $\text{sec} \leq 0$ (Hadamard) | any closed                                 | $(\widehat{X} \setminus X) \cup \bigcup \{\text{defective facets}\} \subsetneq M \setminus X$ |
| $\text{sec} \equiv +1$ ( $\mathbb{S}^n$ )                                 | any closed, $ X  \geq 2$                   | $M \setminus X$ (total)                                                                       |
| compact, $\text{sec} > 0$ (or any compact)                                | $\delta$ -dense, $2\delta < \text{inj } M$ | $M \setminus X$ (total)                                                                       |
| compact, $\text{sec} > 0$                                                 | sparse                                     | total $\Leftarrow$ no cut-shielding (§7)                                                      |

## 2 Setup and notation

Realise  $\mathbb{S}^n \subseteq \mathbb{R}^{n+1}$  as the unit sphere with the round metric; the geodesic distance is

$$d(x, y) = \arccos \langle x, y \rangle \in [0, \pi],$$

and  $B(a, \rho) = \{x : d(x, a) < \rho\}$ ,  $\bar{B}(a, \rho) = \{x : d(x, a) \leq \rho\}$ . For a unit vector  $v$  the *open hemisphere*, *closed hemisphere* and *equator* with pole  $v$  are

$$H_v^\circ = \{\langle \cdot, v \rangle > 0\} = B(v, \frac{\pi}{2}), \quad H_v = \{\langle \cdot, v \rangle \geq 0\} = \bar{B}(v, \frac{\pi}{2}), \quad \partial H_v = \{\langle \cdot, v \rangle = 0\} = S(v, \frac{\pi}{2}).$$

The equator  $\partial H_v$  is the geodesic sphere of radius  $\frac{\pi}{2}$  about  $v$ ; it is a totally geodesic great subsphere, isometric to the round  $\mathbb{S}^{n-1}$  (shape operator  $\Pi = 0$ , so its intrinsic curvature is the ambient  $+1$ ).

Throughout  $X \subseteq \mathbb{S}^n$  is closed (hence compact) and nonempty. We use the medial axis  $M_X$ , reach  $r(a) = d(a, X)$ , and reconstruction  $\mathcal{R}_X = \bigcup_{a \in M_X} B(a, r(a))$  defined in §1. Two elementary remarks are used repeatedly.

*Remark 2.1* (Freeness is antipodal containment). An open ball  $B(a, \rho)$  is *free* ( $B(a, \rho) \cap X = \emptyset$ ) iff  $d(x, a) \geq \rho$  for all  $x \in X$ , i.e.  $\langle x, a \rangle \leq \cos \rho$ , i.e.  $\langle x, -a \rangle \geq \cos(\pi - \rho)$ :

$$B(a, \rho) \text{ free} \iff X \subseteq \bar{B}(-a, \pi - \rho).$$

A medial ball  $B(a, r(a))$  thus corresponds to the *enclosing ball*  $\bar{B}(-a, \pi - r(a))$  of  $X$ , centred at the antipode  $-a$  and touching  $X$  in the feet of  $a$ . In particular  $\mathcal{R}_X \cap X = \emptyset$ , so always  $\mathcal{R}_X \subseteq \mathbb{S}^n \setminus X$ .

*Remark 2.2* (Directional derivative of  $d(\cdot, X)$ ). For  $c \notin X$  let  $\text{Feet}(c) = \{x \in X : d(c, x) = d(c, X)\}$  (nonempty, compact). Writing  $\nu_x \in T_c \mathbb{S}^n$  for the unit initial vector of the geodesic from  $c$  away from  $x$  (so  $\nabla d(\cdot, X)(c) = \nu_x$  when  $c$  is not in the cut locus of  $x$ ), the one-sided derivative of  $d(\cdot, X) = \min_x d(\cdot, x)$  in a direction  $w$  is  $\min_{x \in \text{Feet}(c)} \langle w, \nu_x \rangle$ .

### 3 The focal pole

We first isolate, in its sharpest form, the elementary fact that makes positive curvature different: a free ball that touches  $X$  twice is already medial *at its centre*, and in the hemispherical limit the centre is a circumcentre that swallows everything.

**Lemma 3.1** (Focal pole). *Let  $H_v^\circ$  be a free hemisphere, i.e.  $X \subseteq H_{-v} = \{\langle \cdot, v \rangle \leq 0\}$ , and let  $K_v = X \cap \partial H_v = \{x \in X : \langle x, v \rangle = 0\}$  be its contact set. Then  $d(v, X) = \frac{\pi}{2}$  and  $\text{Feet}(v) = K_v$ . Consequently, if  $|K_v| \geq 2$  then  $v \in M_X$  and*

$$B(v, \frac{\pi}{2}) = H_v^\circ \subseteq \mathcal{R}_X,$$

*the entire hemisphere interior, regardless of the shape of  $K_v$ .*

*Proof.* For  $x \in X$  we have  $\langle x, v \rangle \leq 0$ , i.e.  $d(v, x) = \arccos \langle x, v \rangle \geq \frac{\pi}{2}$ , with equality iff  $\langle x, v \rangle = 0$ , i.e.  $x \in K_v$ . Hence  $d(v, X) = \frac{\pi}{2}$  and the nearest points of  $v$  are exactly  $K_v$ . If  $|K_v| \geq 2$  then  $v$  has two nearest points, so  $v \in M_X$  with  $r(v) = \frac{\pi}{2}$ , and  $B(v, \frac{\pi}{2}) = \{d(\cdot, v) < \frac{\pi}{2}\} = \{\langle \cdot, v \rangle > 0\} = H_v^\circ$ . As  $v \in M_X$ , every point of  $H_v^\circ$  lies in this medial ball, so  $H_v^\circ \subseteq \mathcal{R}_X$ .  $\square$

*Remark 3.2* (Why the Motzkin step disappears). The pole  $v$  is equidistant from the *entire* equator  $\partial H_v = S(v, \frac{\pi}{2})$ , because the equator is a geodesic sphere centred at  $v$ . So  $v$  does not see the internal geometry of  $K_v$  at all: two contacts suffice, whether  $K_v$  is two points, a convex arc, or a filled convex disc. This is the precise sense in which positive curvature *trivialises detection*. In  $\mathbb{R}^n$  the same supporting configuration has its circumcentre at infinity, and a centre at finite

height over a *convex* contact set has a unique foot ( $\text{convex} \Rightarrow \text{Chebyshev}$ ), hence is not medial: this is why the Euclidean theory needs non-convexity. On a sphere of radius  $R$  the pole recedes to distance  $\pi R/2 \rightarrow \infty$  as  $R \rightarrow \infty$  and the focal mechanism degenerates back to the Euclidean one; at any positive curvature the circumcentre is a genuine point and two contacts are enough.

**Proposition 3.3** (Finite equidistant centre — any manifold). *Lemma 3.1 used no property of the round metric. In any Riemannian manifold a free geodesic ball  $B(a, \rho)$  ( $\rho = d(a, X)$ ) has every boundary point at distance  $\rho$  from  $a$ ; so if  $X \cap \partial B(a, \rho)$  has two or more points, then  $a \in M_X$  and  $B(a, \rho) \subseteq \mathcal{R}_X$ , with no condition on the position or geometry of the contacts. A finite equidistant centre never asks “which contact is nearest” — the question whose answer is the Chebyshev structure — so that structure plays no role.*

**Theorem 3.4** (The Busemann sign controls detection). *For a maximal free convex region  $\mathcal{H}$  (the analogue of a horoball), the nature of its centre, and with it the detection problem, is fixed by the sign of  $\text{sec}$  through the Hessian comparison theorem.*

$\text{sec} \leq 0$ :  $\text{Hess } d \succeq$  the flat model, so Busemann functions are convex and  $\mathcal{H} = \{b \leq c\}$  is convex and unbounded, with an ideal centre. A finite centre receding to it sees the contacts at Busemann-distorted distances and is medial iff its foot is non-Chebyshev in  $\partial\mathcal{H}$ : detection is the Chebyshev/Motzkin question (Motzkin on flat horospheres [1], the Carnot/Alexandrov frontier of [4] on curved ones).

$\text{sec} \geq 0$ :  $\text{Hess } d \preceq$  the flat model, so Busemann functions are concave and the convex sublevel sets  $\{b \leq c\}$  cannot grow unbounded: every maximal free convex region is bounded, with a finite centre, and by Proposition 3.3 its defect is exactly  $|X \cap \partial\mathcal{H}| \geq 2$  — the contacts’ geometry idle.

*The boundedness is a theorem, not a heuristic: in the compact case the convexity radius is finite ( $\leq \pi/2\sqrt{\sup \text{sec}}$ ), capping convex balls; in the non-compact case a complete  $\text{sec} > 0$  manifold has a point soul (Cheeger–Gromoll; Perelman), hence no unbounded convex proper subset. Thus the Chebyshev/Motzkin detection layer is a  $\text{sec} \leq 0$  structure with no positive-curvature analogue, on any positively curved space — compact or not, homogeneous or not.*

**Remark 3.5** (No escape hatch; the pincer). Homogeneity does not help: on  $\mathbb{CP}^n$  ( $1 \leq \text{sec} \leq 4$ ) the maximal convex ball has radius the convexity radius  $\frac{\pi}{4}$ , and its finite centre is equidistant from the boundary geodesic sphere — a Berger sphere whose  $U(1)$ -squashing (principal curvatures  $\cot r$  and  $2 \cot 2r$ , here 1 and 0) leaves the equidistance, hence the  $\geq 2$ -contact criterion, untouched. Nor is the intrinsic Berger–Chebyshev question new ground: as  $r \rightarrow \frac{\pi}{2}$  the geodesic sphere collapses onto its  $\mathbb{CP}^{n-1}$  cut locus with the Hopf circle shrinking, degenerating into the sub-Riemannian Heisenberg metric — it is the  $\mathbb{CH}^n$ -horosphere Chebyshev question of [4] in collapsed form, a  $\text{sec} \leq 0$  problem. Nor does non-compactness help: even a paraboloid-type  $\text{sec} > 0$  end has bounded convex regions (point soul), the asymptotically flat regime only *approaching* the  $\text{sec} = 0$  knife-edge (ideal centre, Motzkin) continuously. On the sphere the two exits from the trichotomy form a pincer: drop convexity and the wrap-around balls give *total* reconstruction (Theorem 4.1); keep it and the finite centre gives the  $\geq 2$ -contact criterion — both with no Motzkin step. Lemma 3.1 is the visible symptom of the Busemann sign.

Lemma 3.1 already shows that “most” of  $\mathbb{S}^n$  is reconstructed, with no convexity hypothesis. The next section shows that *all* of  $\mathbb{S}^n \setminus X$  is, by a mechanism that does not even need the hemisphere to be supporting.

## 4 Total reconstruction on the sphere

**Theorem 4.1** (Total reconstruction). *Let  $X \subseteq \mathbb{S}^n$  be closed with  $|X| \geq 2$ . Then*

$$\mathcal{R}_X = \mathbb{S}^n \setminus X.$$

*Proof.* By Remark 2.1,  $\mathcal{R}_X \subseteq \mathbb{S}^n \setminus X$ . We prove the reverse inclusion: every  $p \in \mathbb{S}^n \setminus X$  lies in some open medial ball.

If  $p \in M_X$  then  $p \in B(p, r(p))$  (as  $r(p) = d(p, X) > 0$ ) and we are done. So assume  $p \notin M_X$ :  $p$  has a *unique* nearest point  $x_0 \in X$ , at distance  $s_p := d(p, x_0) = d(p, X) \in (0, \pi)$ . (The distance is  $< \pi$ : if it were  $\pi$  then  $X \subseteq \{-p\}$  would be a single point, contradicting  $|X| \geq 2$ .) In particular  $p \neq -x_0$ , so the minimising geodesic from  $x_0$  to  $p$  is unique; let  $\gamma$  be its unit-speed extension,  $\gamma(0) = x_0$ ,  $\gamma(s_p) = p$ , defined for all  $t \geq 0$  (geodesics on  $\mathbb{S}^n$  are complete). For  $t \in [0, \pi]$  the arc  $\gamma|_{[0, t]}$  is minimising, so  $d(\gamma(t), x_0) = t$ .

Consider

$$t^* = \sup\{t \in [0, \pi] : d(\gamma(t), X) = t\}.$$

On  $[0, \pi]$ ,  $\gamma$  minimises from  $x_0$ , so  $d(\gamma(t), x_0) = t \geq d(\gamma(t), X)$ , with equality on the closed set  $[0, t^*]$  and strict inequality on  $(t^*, \pi]$ . As  $x_0$  is the unique nearest point of  $p = \gamma(s_p)$  and stays nearest on a neighbourhood,  $t^* \geq s_p > 0$ .

*Step 1:*  $t^* < \pi$ . At  $t = \pi$ ,  $\gamma(\pi) = -x_0$ ; since  $|X| \geq 2$  some  $x_1 \neq x_0$  has  $d(-x_0, x_1) < \pi = d(-x_0, x_0)$ , so  $d(-x_0, X) < \pi$  and the supremum is attained before  $\pi$ .

*Step 2:* a free ball through  $p$  with margin  $s_p$ . Put  $c = \gamma(t^*)$ , so  $r(c) = d(c, X) = t^*$  with  $x_0 \in \text{Feet}(c)$ . As  $\gamma|_{[0, t^*]}$  minimises and  $0 < s_p \leq t^* < \pi$ ,

$$d(p, c) = t^* - s_p < t^* = r(c), \quad r(c) - d(p, c) = s_p > 0. \quad (2)$$

So  $p$  lies in the free ball  $B(c, r(c))$  with a definite margin; it remains to make the centre *medial*.

*Step 3:* a medial centre containing  $p$ . If  $c \in M_X$  we are done by (2). This holds for every *finite*  $X$ : a finite set has no focal points, so the singularity of  $d_X$  at  $c$  is a genuine tie between two distinct contacts. In general, however,  $c$  may be a *focal* point of  $X$  carrying the single foot  $x_0$  — a singularity of the *set's* cut locus, distinct from the *point's* cut locus of §7. No first-order test excludes it: by the spherical law of cosines an overtaking contact  $x_t$  ( $t \downarrow t^*$ ,  $\rho_t = d(x_0, x_t)$ ) obeys  $\langle \exp_{x_0}^{-1} x_t, \gamma'(0) \rangle > \frac{1}{2} \rho_t^2 \cot t^* + O(\rho_t^4)$ , whose right side turns *negative* once  $t^* > \frac{\pi}{2}$  — the contact may be displaced backward, away from  $p$  — and  $t^* = \pi - \operatorname{arccot} \kappa_g > \frac{\pi}{2}$  is precisely the focal distance of an outward-convex contact arc of geodesic curvature  $\kappa_g$ . To reach a medial centre regardless, ascend  $d_X = d(\cdot, X)$ , locally semiconcave on  $\mathbb{S}^n \setminus X$  [7]. Lieutier's generalised-gradient flow of  $d_X$  from  $p$  [6] is 1-Lipschitz, increases  $d_X$ , and rests only at its critical points; a critical point  $a \notin X$  has 0 in the convex hull of the unit initial directions of its minimising geodesics to  $X$ , impossible for a single direction, so it has at least two such geodesics. Since  $r(a) = d_X(a) < \pi$  (as  $|X| \geq 2$ ),  $a$  is off the cut locus  $\{-x\}$  of each nearest point  $x$ , so these geodesics reach two *distinct* points and  $a \in M_X$ . On the compact sphere the trajectory has finite length and cannot escape — the sole obstruction in  $\mathbb{R}^n$ , where it runs to infinity with a unique foot (the unreconstructed rays of §1) — so it converges to such an  $a_\infty \in M_X$ . The flow retains its starting centre in the moving free ball (its swallow property [6]): on each unique-foot arc the trajectory is the geodesic from that foot and the margin  $d_X - d(p, \cdot)$  is constant, equal to  $s_p$  on the first arc by (2), the focal crossings costing only higher-order margin. Hence  $p \in B(a_\infty, r(a_\infty)) \subseteq \mathcal{R}_X$ . (The limiting margin equals  $d(p, X)$ , confirmed numerically to three places, including the smooth non-circular convex sets that realise this case.)  $\square$



*Remark 4.2* (Compactness replaces the escape to infinity). The only place the argument can fail in  $\mathbb{R}^n$  is Step 1: there the ray  $\gamma$  leaves to infinity and  $x_0$  may remain the unique nearest point forever (the “rays past the segment” of §1), so  $t^* = +\infty$  and no medial centre is produced. On  $\mathbb{S}^n$  the ray returns: at the antipode  $x_0$  is maximally far, so it is overtaken strictly earlier and  $t^* < \pi$ . The margin  $r(c) - d(p, c) = s_p = d(p, X)$  is exactly the depth of  $p$  below the moving free sphere, the constant-margin form of the curvature-free swallow of [4, Lem. 5.5], here with the moving centre  $\gamma(t)$  reaching a genuine medial point rather than drifting to an ideal one.

*Remark 4.3* (Hypotheses are minimal). No convexity, position (“ $X$  in a hemisphere”), or regularity hypothesis is used;  $X$  need not be a hypersurface or graph-type, in contrast to [2, 3, 4]. The only hypothesis is  $|X| \geq 2$ , and it is necessary: for a single point  $x_0$  one has  $M_X = \emptyset$  and  $\mathcal{R}_X = \emptyset \neq \mathbb{S}^n \setminus \{x_0\}$ . The focal pole (Lemma 3.1) is the special case in which the medial centre  $c$  produced by the proof is a pole ( $t^* = \frac{\pi}{2}$ ); the theorem shows the same conclusion with  $c$  an arbitrary medial centre, which is why no supporting hemisphere is needed.

*Remark 4.4* (Numerical confirmation). The construction of the proof was run on  $\mathbb{S}^2$  for 400 random  $X$  (2 to 6 points) and random  $p \notin X$ : in every case the first overtaking parameter  $t^*$  produced a centre  $c$  with  $\geq 2$  feet and  $d(p, c) < r(c)$ , with minimal observed margin 0.028 (all margins  $= d(p, X) > 0$ , as predicted). The degenerate witnesses of §5 — the back-arc of two points and the southern points of a latitude circle — were confirmed reconstructed.

## 5 Collapse of the Białożyty structure

Theorem 4.1 already shows the Euclidean certificate (1) cannot transfer verbatim: its right-hand side is in general a proper subset of  $\mathbb{S}^n \setminus X$ . We record two explicit witnesses, because they pinpoint *which* part of the certificate fails. Write  $\widehat{X}$  for the spherical convex hull (the intersection of the closed hemispheres containing  $X$ ); for  $X$  in an open hemisphere this is the smallest geodesically convex closed set containing  $X$ . The naive analogue of (1) would read

$$\mathcal{R}_X \stackrel{?}{=} (\widehat{X} \setminus X) \cup \bigcup \{H_v^\circ : H_v^\circ \text{ free supporting, } X \cap \partial H_v \text{ non-convex in } \partial H_v\}. \quad (3)$$

**Proposition 5.1** (A convex set reconstructs its whole complement). *Let  $X = \{x : \langle x, e \rangle \geq \cos \beta\} = \bar{B}(e, \beta)$  be a closed spherical cap of radius  $\beta \in (0, \frac{\pi}{2})$  about a pole  $e$  (geodesically convex). Then  $\mathcal{R}_X = \mathbb{S}^n \setminus X$ , whereas the right-hand side of (3) is empty.*

*Proof.*  $X$  is convex, so  $\widehat{X} = X$  and  $\widehat{X} \setminus X = \emptyset$ ; and a supporting equator of a convex set with smooth strictly convex boundary touches it in a single point, so there is no defective (non-convex contact) supporting hemisphere. Thus the right-hand side of (3) is empty. But  $X = \bar{B}(e, \beta)$  has more than two points, so Theorem 4.1 gives  $\mathcal{R}_X = \mathbb{S}^n \setminus X$ . Concretely, the antipode  $-e$  has  $d(-e, x) = \pi - \arccos \langle x, e \rangle \geq \pi - \beta$  for all  $x \in X$ , with equality on the boundary circle  $\{\langle \cdot, e \rangle = \cos \beta\}$ ; so  $-e \in M_X$  with  $r(-e) = \pi - \beta$  and  $B(-e, \pi - \beta) = \{\langle \cdot, e \rangle < \cos \beta\} = \mathbb{S}^n \setminus X$  already reconstructs the entire complement from a single focal centre.  $\square$

**Proposition 5.2** (A circle whose far hemisphere is reconstructed). *Let  $X = \{\langle \cdot, e \rangle = \cos \beta\}$  be a latitude circle of colatitude  $\beta \in (0, \frac{\pi}{2})$  on  $\mathbb{S}^2$ . Then  $\mathcal{R}_X = \mathbb{S}^2 \setminus X$ , while the right-hand side of (3) is the open cap  $\{\langle \cdot, e \rangle > \cos \beta\} = \widehat{X} \setminus X$  only — it omits the entire southern region  $\{\langle \cdot, e \rangle < \cos \beta\}$ .*

*Proof.*  $\widehat{X}$  is the closed cap  $\bar{B}(e, \beta)$ , so  $\widehat{X} \setminus X$  is the open cap. A great circle supporting the latitude circle is tangent to it, meeting it in one point, so every supporting hemisphere has a single (hence convex) contact: no defective hemisphere exists and the union in (3) is empty. Thus the right-hand side of (3) is the open cap. Yet  $e$  (centre of the circle, all of  $X$  at distance  $\beta$ ) and  $-e$  (all of  $X$  at distance  $\pi - \beta$ ) are both medial, with  $B(e, \beta) = \{\langle \cdot, e \rangle > \cos \beta\}$  and  $B(-e, \pi - \beta) = \{\langle \cdot, e \rangle < \cos \beta\}$ , whose union is  $\mathbb{S}^2 \setminus X$  (consistent with Theorem 4.1). The southern region is reconstructed by the focal centre  $-e$ , invisible to (3).  $\square$

*Remark 5.3* (What the certificate was, and is not). Propositions 5.1–5.2 locate the failure of (3) precisely in its *second* term. The hull term  $\widehat{X} \setminus X$  remains a *subset* of  $\mathcal{R}_X$  (it is reconstructed — e.g. by the centre  $e$  in both examples), but the “defective supporting hemispheres” fail to capture the reconstruction coming from the *interior focal centres* ( $-e$  above), which have no boundary contact structure to be “defective”. In  $\mathbb{R}^n$  these focal centres sit at infinity and contribute nothing; on  $\mathbb{S}^n$  they are ordinary medial points and contribute most of  $\mathcal{R}_X$ . The organising notion of the non-positively curved theory — a defect read off a contact set — has therefore no positive-curvature counterpart: there is nothing to detect, because everything off  $X$  is reconstructed.

## 6 Generalisation: compact manifolds and the spread condition

The proof of Theorem 4.1 used the round geometry twice: in Step 1, where a rival (or focal turn) appears before the antipode, and in Step 3, where every critical point of  $d_X$  has reach  $< \pi$  and hence two *distinct* nearest points. On a general compact manifold both are secured by a single spread hypothesis on  $X$ , which caps the reach below  $\text{inj}(M)$ . Recall  $\text{inj}(M) > 0$  for compact  $M$ , and that a geodesic is minimising on every sub-arc of length  $< \text{inj}(M)$ .

**Theorem 6.1** (Total reconstruction under a spread condition). *Let  $M$  be a compact Riemannian manifold and  $X \subseteq M$  closed and  $\delta$ -dense (every point of  $M$  is within  $\delta$  of  $X$ ). If*

$$2\delta < \text{inj}(M),$$

*then  $\mathcal{R}_X = M \setminus X$ .*

*Proof.*  $\mathcal{R}_X \subseteq M \setminus X$  as the balls are free. Fix  $p \in M \setminus X$ ; if  $p \in M_X$  then  $p \in B(p, r(p)) \subseteq \mathcal{R}_X$ . Otherwise ascend  $d_X = d(\cdot, X)$  from  $p$  by its semiconcave generalised-gradient flow (Step 3 of Theorem 4.1): on the compact  $M$  the trajectory has finite length and converges to a critical point  $a_\infty$  of  $d_X$ , retaining  $p$  in the moving free ball, so  $d(p, a_\infty) < r(a_\infty) = d_X(a_\infty)$ . By  $\delta$ -density  $d_X \leq \delta$  on all of  $M$ , whence  $r(a_\infty) \leq \delta < \frac{1}{2} \text{inj}(M)$ ; thus  $a_\infty$  lies off the cut locus of each of its nearest points, its  $\geq 2$  minimising geodesics (it is a critical point) reach  $\geq 2$  *distinct* points, and  $a_\infty \in M_X$ . Hence  $p \in B(a_\infty, r(a_\infty)) \subseteq \mathcal{R}_X$ . The density bound is what forces every critical point below the cut locus, turning the two-geodesic centre the flow finds into a two-*point* one.  $\square$

*Remark 6.2* (Positive curvature: making the hypothesis explicit). Theorem 6.1 is curvature-free, but a curvature upper bound turns it into an effective statement, which is the natural positive-curvature reading. By Klingenberg’s lemma, for compact  $M$ ,

$$\text{inj}(M) \geq \min \left\{ \frac{\pi}{\sqrt{\Delta}}, \frac{1}{2} \ell(M) \right\} \quad (\text{sec} \leq \Delta),$$



where  $\ell(M)$  is the length of the shortest closed geodesic and  $\frac{\pi}{\sqrt{\Delta}}$  is the conjugate radius bound (Rauch). Thus if  $\kappa \leq \sec \leq \Delta$  and  $X$  is  $\delta$ -dense with

$$2\delta < \min\left\{\frac{\pi}{\sqrt{\Delta}}, \frac{1}{2}\ell(M)\right\},$$

then  $\mathcal{R}_X = M \setminus X$ . For  $M = \mathbb{S}^n$  ( $\kappa = \Delta = 1$ ,  $\ell = 2\pi$ ) this gives total reconstruction once  $X$  is  $\frac{\pi}{2}$ -dense; but Theorem 4.1 shows that on  $\mathbb{S}^n$  the density hypothesis is in fact *unnecessary*. That is the content of the next remark: constant curvature is precisely where the spread condition can be dropped.

*Remark 6.3* (Why constant curvature needs no spread). In Theorem 6.1 the spread hypothesis caps the reach,  $d_X \leq \delta < \text{inj}(M)$ , placing every critical point of  $d_X$  below the cut locus, where its  $\geq 2$  minimising geodesics reach *distinct* points. On  $\mathbb{S}^n$  this is automatic for *any*  $X$  with  $|X| \geq 2$ , with no spread: the cut locus of each point is the single antipode at the *diameter*  $\pi$ , so a critical point (reach  $< \pi$ , as  $|X| \geq 2$ ) already has two distinct nearest points — equivalently, along a nearest-point ray the antipode is where  $x_0$  becomes the globally farthest contact, so a rival overtakes it first (Step 1 of Theorem 4.1). The two italicised features — the cut point realises the diameter, and it does so in every direction simultaneously — are exactly the rigidity of constant positive curvature. They fail in variable positive curvature, which is the subject of §7.

## 7 The variable-curvature obstruction: cut shielding

Off constant curvature, the conclusion  $\mathcal{R}_X = M \setminus X$  can fail for sparse  $X$ . The obstruction is structural and worth stating precisely, even though we do not have a closed-form characterisation in the variable case: it is the only gap between Theorem 6.1 and the unconditional Theorem 4.1.

The proof of Theorem 4.1 produces a medial centre by following  $\gamma$  from  $x_0$  until a *second contact* appears. If instead  $\gamma$  reaches the *cut locus* of  $x_0$  first — a point  $m = \gamma(\ell)$  admitting a second minimising geodesic back to the same  $x_0$ , with no rival contact of  $X$  yet within distance  $\ell$  — then  $m$  is not medial ( $x_0$  is still its unique nearest *point* of  $X$ , merely reached twice), the margin identity breaks past  $m$  (there  $d(\gamma(t), x_0) < t$ ), and the construction stalls. We call such a configuration *cut-shielding* of the points  $\gamma(s)$ ,  $s < \ell$ .

**Proposition 7.1** (The margin argument fails exactly at cut shielding). *Let  $p \notin X$  have unique nearest point  $x_0$ , and let  $\ell = \text{cut}(x_0; \gamma)$  be the distance to the cut locus of  $x_0$  along the ray  $\gamma$  through  $p$ . If no point of  $X \setminus \{x_0\}$  lies within distance  $\ell$  of any  $\gamma(t)$ ,  $t \leq \ell$  (i.e.  $x_0$  is the unique nearest point of  $X$  all along  $\gamma|_{[0, \ell]}$ ), then the moving-centre construction of Theorems 4.1–6.1 produces no medial centre on  $\gamma$ .*

*Proof.* For  $t \leq \ell$  the arc is minimising and  $x_0$  is the unique nearest point, so  $d(\gamma(t), X) = t$  with  $\text{Feet}(\gamma(t)) = \{x_0\}$ : no  $\gamma(t)$ ,  $t \leq \ell$ , is medial. At  $t = \ell$  the point  $m = \gamma(\ell)$  has a second minimising geodesic to  $x_0$ , but its unique nearest *point* of  $X$  is still  $x_0$ , so  $m \notin M_X$ . For  $t > \ell$ ,  $d(\gamma(t), x_0) < t$ , so the identity  $d(\gamma(t), X) = t$  fails and the constant margin  $r - d(p, \cdot) = s_p$  is no longer maintained.  $\square$

*Remark 7.2* (A concrete model, and the honest status). The mechanism is visible on a prolate ellipsoid of revolution (a “cigar”,  $\sec > 0$  everywhere) with a short waist closed geodesic of length  $\ell_0$ . Take  $X = \{x_0, x_1\}$  with  $x_0$  on the waist and  $x_1$  at the far tip. The waist geodesic through  $x_0$

returns to  $x_0$  along two arcs of length  $\ell_0/2$ ; the diametrically opposite waist point  $m$  is a cut point of  $x_0$  with  $x_0$  still its unique nearest contact (the tip  $x_1$  being far). By Proposition 7.1 the waist points between  $x_0$  and  $m$  are not reconstructed by waist centres; whether some *off*-waist medial centre nonetheless reconstructs them is decided by the *global* two-point medial axis, which for  $|X| = 2$  is exactly the bisector  $\{a : d(a, x_0) = d(a, x_1)\}$ , so  $q \in \mathcal{R}_X$  iff some bisector point  $a$  has  $d(a, q) < d(a, x_0)$ . We have now settled this finite test numerically, and the answer is *negative*: cut-shielding genuinely obstructs two-point reconstruction once the curvature varies enough.

- *Unconditional totality*: constant curvature  $+1$  (Theorem 4.1); and any compact  $M$  with  $\delta$ -dense  $X$ ,  $2\delta < \text{inj } M$  (Theorem 6.1). These cover all *sufficiently sampled*  $X$  in positive curvature.
- *Settled (numerically, negative) for sparse  $X$* : computing the bisector test above on a prolate ellipsoid of revolution (the cigar;  $\text{sec} > 0$  everywhere), with the whole pipeline validated on the round sphere (geodesic distances within 1%; reconstruction total, zero shielded points, confirming Theorem 4.1), gives a sharp transition in the eccentricity. Reconstruction is total for small eccentricity and shielded for large, with the transition localised to  $c^* = 2.6 \pm 0.1$  (the 2.4% solver error cannot resolve it more finely); the shielded region sits near the far tip. At  $c = 3$  it is  $\approx 9\%$  of the surface with reconstruction deficit  $\min_a [d(a, q) - r(a)] \approx 18\%$  of the diameter at its deepest point (the distance by which  $q$  lies beyond every medial ball), a factor of about seven above the 2.4% discretisation error, and persisting and stabilising under mesh refinement (its area fraction rising slightly, not vanishing). So  $\mathcal{R}_X = M \setminus X$  *fails* for sparse  $X$  in variable positive curvature: total two-point reconstruction is special to constant curvature (and, more finely, bounded eccentricity). The remaining open question is only the *quantitative* threshold — precisely,  $\text{cut}(x_0; \gamma) \geq \sup\{t : x_0 \text{ uniquely nearest}\}$  along every ray, which on  $\mathbb{S}^n$  holds because the cut distance  $\pi$  realises the diameter (Remark 6.3) and which the eccentricity controls on the cigar.

The shielded points are cut points of  $x_0$  admitting two or more minimising geodesics to  $x_0$  yet strictly closer to  $x_0$  than to  $x_1$  (a single nearest *point*, reached twice), hence lie in  $M_X^g \setminus M_X$  and *are* reconstructed by the two-geodesic axis of §8 (Theorem 8.3): cut-shielding is the exact phenomenon separating the two axes off constant curvature, and the two-geodesic reformulation is its unconditional resolution.

*Remark 7.3* (Constant-curvature space forms). For a spherical space form  $\mathbb{S}^n/\Gamma$  ( $\text{sec} \equiv +1$ , compact) the local geometry is that of  $\mathbb{S}^n$  but the cut locus of a point is no longer a single antipode (e.g.  $\mathbb{RP}^n$  has diameter  $\frac{\pi}{2}$  and cut locus a hyperplane  $\mathbb{RP}^{n-1}$ ). Theorem 6.1 applies whenever  $X$  is  $\delta$ -dense with  $2\delta < \text{inj}(\mathbb{S}^n/\Gamma)$ ; for sparse  $X$  the cut locus is again the only obstruction, now genuinely nontrivial, and we leave the sharp statement open. Thus even within constant curvature it is the *simple connectivity* of  $\mathbb{S}^n$  — forcing the cut locus to a single diameter-realising point — that makes total reconstruction unconditional.

## 8 The two-geodesic medial axis

The cut-shielding obstruction of §7 is an artefact of *which* medial axis one reconstructs from. Białożył's  $M_X$  counts points with two distinct nearest *points* of  $X$ . A shielded point  $m = \gamma(\ell)$  has a single nearest point  $x_0$  reached by two distinct minimising *geodesics* — it is invisible to  $M_X$  but is exactly the kind of point the cut locus is made of. Reconstructing instead from the

two-geodesic medial axis removes the obstruction altogether, and does so without disturbing the sphere or the Euclidean theory.

**Definition 8.1** (Two-geodesic medial axis). *For  $a \in M \setminus X$  let  $r(a) = d(a, X)$ . Say  $a \in M_X^g$  if there are two distinct minimising geodesics from  $a$  to  $X$  (their endpoints in  $X$  may coincide). Put  $\mathcal{R}_X^g = \bigcup_{a \in M_X^g} B(a, r(a))$ . Since two distinct nearest points give two distinct geodesics,  $M_X \subseteq M_X^g$ , hence  $\mathcal{R}_X \subseteq \mathcal{R}_X^g \subseteq M \setminus X$ ; and  $M_X^g$  is (the two-sided part of) the singular set of  $d(\cdot, X)$  — the cut locus of  $X$ .*

**Proposition 8.2** (The change is invisible in  $\mathbb{R}^n$  and on  $\mathbb{S}^n$ ). *If  $M = \mathbb{R}^n$ , or  $M = \mathbb{S}^n$  with  $|X| \geq 2$ , then  $M_X^g = M_X$  and  $\mathcal{R}_X^g = \mathcal{R}_X$ .*

*Proof.* In  $\mathbb{R}^n$  any two points are joined by a unique geodesic, so two distinct minimising geodesics to  $X$  have distinct endpoints, i.e. two nearest points. On  $\mathbb{S}^n$ , two distinct minimising geodesics from  $a$  to a single  $x_0 \in X$  force  $a$  into the cut locus of  $x_0$ , which is the antipode  $\{-x_0\}$  at distance  $\pi$ ; then  $r(a) = d(a, X) = \pi$  makes  $x_0$  the unique point of  $X$ , contradicting  $|X| \geq 2$ . So in both cases distinct geodesics have distinct endpoints,  $M_X^g = M_X$ .  $\square$

Thus Theorem 4.1 (total on  $\mathbb{S}^n$ ) and Białożył’s theorem (the defect structure in  $\mathbb{R}^n$ ) are unchanged by the new definition. The definitions part company only where there is a cut locus — variable curvature, or nontrivial topology — which is exactly where the new one pays off.

**Theorem 8.3** (Total reconstruction from the two-geodesic axis). *Let  $M$  be any compact Riemannian manifold. Then for every closed  $X \subseteq M$  with  $|X| \geq 2$ ,*

$$\mathcal{R}_X^g = M \setminus X.$$

*No spread or curvature hypothesis is needed. The containment of  $p$  in the moving ball past intermediate cut points — the one formerly delicate step — is proved unconditionally in the final note of the series [5, Thm. 3.1, Cor. 3.2]; Remark 8.4 records the scope of the elementary margin argument and the repair.*

*Proof.*  $\mathcal{R}_X^g \subseteq M \setminus X$  as the balls are free. Fix  $p \in M \setminus X$ ; if  $p \in M_X^g$  then  $p \in B(p, r(p)) \subseteq \mathcal{R}_X^g$ . Otherwise ascend the semiconcave  $d_X$  from  $p$  (Step 3 of Theorem 4.1): on the compact  $M$  the trajectory has finite length and converges to a critical point  $a_\infty$  of  $d_X$ , retaining  $p$  in the moving free ball. A critical point has 0 in the convex hull of the unit initial directions of its minimising geodesics to  $X$ , hence *two or more* such geodesics — that is,  $a_\infty \in M_X^g$ , with no need to decide whether their endpoints coincide (which is exactly the focal/conjugate subtlety that the two-point axis must confront and the geodesic axis sidesteps). Thus  $p \in B(a_\infty, r(a_\infty)) \subseteq \mathcal{R}_X^g$ .  $\square$

*Remark 8.4* (Scope of the containment step). The clause “retaining  $p$  in the moving free ball” is Lieutier’s swallow property [6], proved in  $\mathbb{R}^n$ . In the elementary form of Step 3 the margin  $r - d(p, \cdot)$  is constant along each unique-foot arc *because that arc is the minimising geodesic from its foot* — which holds exactly while the reach stays below the foot’s cut distance: unconditional on  $\mathbb{S}^n$  (cut distance  $\pi = \text{diameter}$ ) and under the spread hypothesis  $2\delta < \text{inj } M$  of Theorem 6.1. On a general compact  $M$  the reach can exceed an intermediate foot’s cut distance, where the arc ceases to minimise and the constant-margin identity breaks. This gap is now closed: the final note of the series proves the containment unconditionally on every complete manifold, with no genericity or analyticity hypothesis, by following the ascent only while the minimising geodesic to  $X$  is unique — there  $d_X$  grows at unit rate and the margin cannot shrink, through focal

and conjugate points without comment — and *stopping at the first approach* to  $M_X^g$ , where the accumulated margin is intact, instead of flowing through it [5, Thm. 3.1, Cor. 3.2]. Theorem 8.3 therefore holds as stated for every compact  $M$ , and indeed for every closed  $X \neq \emptyset$ : even  $|X| = 1$ , invisible to the two-point axis, is reconstructed in full by the two-geodesic one.

*Remark 8.5* (The fix activates exactly on the cut locus). The flow never needs a rival contact: when none appears it simply rests at a cut point of  $x_0$  — two minimising geodesics to a single point — which the two-geodesic axis counts as medial. This is precisely the shielded point  $m$  of §7, now medial. The difference between the two theories,

$$M_X^g \setminus M_X = \{\text{points with a unique nearest point of } X \text{ but two minimising geodesics to it}\},$$

is empty in  $\mathbb{R}^n$  and on  $\mathbb{S}^n$  (Proposition 8.2) and carries the entire repair in variable curvature. So total reconstruction, which for  $M_X$  is special to constant curvature (Remark 6.3), holds for  $M_X^g$  on *every* compact manifold — positive curvature is no longer distinguished, because the mechanism is now compactness of the cut locus rather than the antipodal rigidity of the round metric.

*Remark 8.6* (Which medial axis, and a caveat). The two axes answer different questions.  $M_X$  — points with two nearest *features* — is the symmetry/branching set, the shape skeleton of computational geometry;  $M_X^g$  — the singular set of  $d(\cdot, X)$ , i.e. the cut locus of  $X$  — is the natural object of the distance-function viewpoint [6] and of the Riemannian cut-locus theory. For the *reconstruction* question “is every point off  $X$  captured?” the geodesic axis is the clean choice, trivialising it on any compact manifold; for the *structural* question “what does the skeleton look like?” the point axis is the informative one, since  $M_X^g$  conflates a genuine branch of  $X$  with two routes to a single feature. Note that Theorem 8.3 needs no genericity: the gradient flow lands at a critical point, which carries two minimising geodesics by definition, so the focal/conjugate cut points that obstruct the two-point argument are absorbed without comment.

## 9 Discussion

**The phenomenon, in one line.** In non-positive curvature a point escapes reconstruction by sending its nearest-point ray to infinity with a single foot; medial reconstruction is therefore partial and is organised by a convexity defect of contact sets (Białożył, [1, 2, 3, 4]). On the sphere geodesics are compact and the ray cannot escape: a second foot is forced before the antipode, every point is swallowed, and  $\mathcal{R}_X = \mathbb{S}^n \setminus X$ . Positive curvature thus removes the reconstruction obstruction *entirely*, the exact opposite of the expectation that a positively curved horosphere is the “hardest” case for detection — there is nothing left to detect.

**Relation to the horospherical programme.** In the Hadamard framework of [4] the ambient ideal boundary indexes the maximal free horoballs, and reconstruction outside the hull is drift to an ideal centre. On  $\mathbb{S}^n$  the ideal boundary is *internalised*: the role of an ideal point is played by a genuine pole  $v$ , the Busemann function by the signed distance  $d(\cdot, v) - \frac{\pi}{2}$  to the equator, and “drift to  $\xi$ ” by convergence of the moving centre  $\gamma(t)$  to a medial point at finite distance. The curvature-free swallow lemma of [4] survives verbatim (Remark 4.2); what does *not* survive is the backbone that made the swallow’s hypothesis (a drifting medial centre) generic only at defective horoballs — on  $\mathbb{S}^n$  every direction supplies one, so the swallow runs everywhere.

**Where nontrivial positive-curvature structure lives.** Total reconstruction makes the *set* reconstruction problem trivial on the sphere, which suggests the informative positive-curvature question is the *domain* reconstruction problem of Lieutier [6]: for an open  $\Omega \subseteq \mathbb{S}^n$ , reconstruct  $\Omega$  from the medial axis of  $\partial\Omega$  using only balls *contained in*  $\Omega$ . There the focal centres on the far side of  $\partial\Omega$  are excluded by the confinement to  $\Omega$ , the antipodal collapse does not occur, and a genuine convexity/defect theory — now with the spherical, positively curved induced geometry of  $\partial\Omega$  — should re-emerge. That is the natural sequel.

## Summary.

| Statement                                                                | Setting                                        | Status                |
|--------------------------------------------------------------------------|------------------------------------------------|-----------------------|
| Focal pole: $ K_v  \geq 2 \Rightarrow H_v^\circ \subseteq \mathcal{R}_X$ | $\mathbb{S}^n$                                 | Lem. 3.1              |
| <b>Total reconstruction</b> $\mathcal{R}_X = \mathbb{S}^n \setminus X$   | $\mathbb{S}^n$ , any closed $X$ , $ X  \geq 2$ | <b>Thm. 4.1</b>       |
| Białożył certificate strictly undercounts                                | $\mathbb{S}^n$ (cap, circle)                   | Prop. 5.1, 5.2        |
| Total reconstruction under spread $2\delta < \text{inj } M$              | any compact $M$                                | Thm. 6.1              |
| Explicit via $\text{sec} \leq \Delta$ (Klingenberg)                      | compact, $\text{sec} \leq \Delta$              | Rem. 6.2              |
| Cut-shielding obstruction; sparse variable case                          | compact $\text{sec} > 0$                       | open (Prop. 7.1)      |
| <b>Two-geodesic axis</b> $\mathcal{R}_X^g = M \setminus X$ (no spread)   | any compact $M$                                | <b>Thm. 8.3</b> , [5] |
| $M_X^g = M_X$ (definition invisible)                                     | $\mathbb{R}^n, \mathbb{S}^n$                   | Prop. 8.2             |

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